

AIDAN M. MOKALLA

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UNDERGRADUATE EDUCATION

Reed College, Portland, Oregon, USA,

2025

+ Spanish minor, sole member of graduating class with double majors: Computer Science & Linguistics,

+ A grades in Real Analysis, Linear Algebra, Deep Learning, Linguistic Field Methods, and Psycholinguistics,

Bryn Mawr College, Dushanbe, Tajikistan,

2024

+ Fellowship funded by international non-profit American Councils to research Persian language varieties,

La Universidad de Carlos III and Middlebury College, Madrid, Spain,

2023

+ Spanish-language courses including Artificial Intelligence, Data Science, and International IP Law.

SELECTED PUBLICATION

Source of the Storm: Interpreting Weather Models to Detect Cyclogenesis,

forthcoming

+ Applying post-superposition statistical interpretation methods to weather prediction models to improve localization of rare climatic events and demonstrate methodological generalizability beyond transformer-based models,

Evaluating Diffusion Language Models Beyond Benchmarks

COLM 2026

+ Found that people, when evaluating the outputs of the recently invented diffusion-based LLM architecture, consistently over-trust AI outputs at a significantly lower rate than when evaluating outputs of traditional LLMs,

Interpretable Signals for VLA Model Failure,

ICML 2026

+ Discovered representational mechanistic predictors of vision-language-action model task failure,

Fault- and Network-Agnostic Agreement, with Erica Blum,

TCC 2026

+ Proved tight running-time bounds for all algorithms achieving network-agnostic Byzantine consensus,

Language-Agnostic Concept Spaces Found in Large Language Models, with Greg Anderson, thesis, 2025

+ Designed, implemented, and evaluated novel architecture to interpret transformer-based LLMs by leveraging natural-language translation of ground-truth semantics. My approach learns convergent “concept spaces” over LLM activations in parallel, yielding more consistent LLM interpretations than prior work,

Modulated Signal Filtering: Wouldn't it be Noise?, with Sameer ud Dowla Khan,

thesis, 2025

+ Collected over 3,000 participant responses to investigate how listeners use unique cognitive strategies to parse spoken speech versus to parse song lyrics, using both deep and symbolic statistical methods,

+ Demonstrated that we use contextually-differentiated mechanisms to filter noise from the “linguistic signal” by making significantly more accurate predictions of how people interpret unclear speech under this prior.

LEADERSHIP & EXPERIENCE

Algoverse AI Research, Machine Learning Researcher, Fellow, & Mentor

September 2025 – present

+ Investigating and publishing methods of explainable AI and climate science,

Linguistic Diversity Initiative, Lead Volunteer Ambassador,

June 2023 – August 2025

+ Leading a team of three translators, I compiled and published dictionaries of terminology used in Portland Public’s science curricula, translated into 6 languages, for households with low English proficiency, already enabling over 800 families to engage in their students’ education,

+ Taught seminars to students and faculty about sociolinguistic research on language use in undergraduate settings,

+ Organized and lead twice-monthly meetings of over 50 volunteers and larger events,

Reed College Student Body, Senator & Honor Councillor,

September 2021 – May 2025

+ Peer-elected to serve in faculty committees including the Committee on Academic Policies and Planning and Committee on Computer Policy and Planning, disburse 6-figure Student Body Fund and other resources to college organizations, and interview and hire all student applicants for on-campus roles,

+ Mediated and resolved conflicts between students, professors, and other members of the college community,

Stoel Rives, LLC, Intern Practise Assistant,

May – August 2022

+ Implemented compliant automation of docketing, court appearances, and other events from firm-wide databases,

+ Maintained communication on behalf of attorneys while drafting and proofreading attorney correspondences.

SKILLS & ACCOLADES

Computational: Python, various data and NLP pipelines, PyTorch, NumPy, JS, Next.js, C++, PHP, SQL, Swift,

Awards: Presidential Commendation for Academic Excellence, Reed College Music Scholarship, National Diversity in STEM Grant, Department of State Critical Languages Scholarship,

Certifications: Federally eligible for non-competitive civil service, licensed in Conflict Mediation & Resolution, CPR,

Languages: Fluent Spanish, English, intermediate Farsi, Tajik, French, basic Portuguese, Dari, Chinese (Mandarin).